

Designing for Human Data Interaction in Data-Driven Media Experiences

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ABSTRACT

The explication of socio-technical challenges like privacy concerns, data transparency and control, accountability issues and lack of trust, posed by personal data leverage in media research has been of interest recently. While many of these challenges have been studied and often discussed in various disciplines ranging from policy to technology to sociology, effective responses that alleviate them are yet to be realised. This calls for holistic approaches that enable inter-disciplinary perspectives and methodologies that speak to these wider challenges in more comprehensive and thus, effective manners. Human Data Interaction, such a cross-disciplinary branch of knowledge, inspired by HCI, seeks to alleviate said challenges through three overarching principles : data legibility, negotiability and agency. But, given the emergent nature of this ‘fledgeling’ domain, the practical realization of these theoretical tenets are yet to be operationalized. This paper reports the design of one such initiative that integrates the principles of HDI, supported by wider research, into the design of a novel, data driven media experience : a Cross Media Profiler. The first half of the paper presents the socio-technical challenges confronting the turn to personal data use in media experiences, rationalizes the need and scope for inter-disciplinary approaches like HDI here and reviews wider literature that is in line with the principles of HDI. The rest of the paper reports the design of the CMP, particularly the realization of the HDI principles within this particular media service proposition. We intend for this design intervention to be leveraged in formative and evaluative studies that test user response to the design decisions made and their effectiveness in alleviating socio-technical challenges while also contributing to the current knowledge base supporting HDI.

CCS CONCEPTS

• **Human-centered computing;**

KEYWORDS

Media Experience, Personal Data, Human Data Interaction

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1 INTRODUCTION

As innovation in media experiences increasingly turn towards personal data leverage that enables more opportunities for content relevance and audience engagement, this shift introduces a number of socio-technical challenges such as trust concerns around black box algorithms and bias, accountability issues on legal, computational and social levels, privacy fears and challenges around transparency and control [32]. While these topics are much debated in disciplines ranging from digital economy [20] to ethics [28, 31] to computer science [37] and are now often seemingly classified as ‘obvious’, their prevalence and impact continue to persist. Hence, these challenges, which are often studied in siloed and disparate manners call for holistic and practically realisable responses in this rapidly changing, emergent context. Here, HCI methods which are inherently interdisciplinary in ethos, approach and practice, offer a means to account for the multidisciplinary nature of these pressing challenges and further explore and respond to these challenges in practice.

An interdisciplinary branch of knowledge, inspired by HCI, that looks into the various aspects of these challenges while proposing focused responses, is Human Data Interaction (HDI) [25]. HDI derives knowledge from multiple disciplines ranging from law to economics to sociology, to devise data sensitive responses through three core values: data legibility, negotiability and agency. But for these theoretical principles to be effective in alleviating challenges in the real world, they require contextualised application, study and unpacking which informs practical design strategies that respond to these challenges. HCI, with its tradition of user-centric and applied research methods stands in the unique position to accommodate these multiple theoretical perspectives while allowing for their contextualised practical design, application and study in real world contexts.

This paper primarily reports the design and inclusion of practical, data sensitive responses aimed at alleviating the socio-technical challenges of personal data use in media experiences. The first half of the paper presents a detailed explication of these socio-technical challenges particularly contextualised within media experiences during its turn towards personal data leverage [34, 35]. It highlights the interdisciplinary nature of these challenges and the need to move from siloed, disparate response measures to more holistic and

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cross-disciplinary interventions. Here, we introduce Human Data Interaction, a novel multi-disciplinary approach towards responding to said challenges, inspired by HCI. The three tenets of HDI : data legibility, negotiability and agency are discussed here and satellite research which associate with these principles are reviewed while scoping out gaps and potential for future HCI intervention for furthering of this agenda. The rest of the paper is dedicated to an account of how our own understandings of the theoretical principles of HDI - data legibility, negotiability and agency, supported by wider research and design recommendations, are operationalized through design in a novel data driven media experience prototype : a Cross Media Profiler. This CMP combines a personal data store - an edge device that supports data portability along with a media profiler and a basic recommendation engine. The service would enable a user to; build a personal media profile by accessing and retrieving media profiles from multiple digital media services; review and modify this profile and then use it within the service to retrieve content recommendations. The prototype is intended to be used to both test user response to our design interventions aimed at alleviating socio-technical challenges of personal data use by media experiences and extend current understandings of HDI as it relates to media experiences.

2 CHALLENGES OF THE TURN TO PERSONAL DATA IN MEDIA EXPERIENCES

As media innovation turns towards personal data leverage : from media recommendations [26, 30] to international broadcasters actively investing in new data-driven technologies for the delivery of real-time, context aware media [3, 9, 38], this shift embodies a number of inherent socio-technical challenges. We present these challenges and responses to them, contextualised within HCI. While a comprehensive review of this literature demands a full publication unto itself, we use the limited scope of the following discussion to highlight the challenges pertinent to the context of media experiences and highlight those responses that inspired design decisions in the intervention presented in the sections that follow.

A key challenge facing the turn to personal data in media is Privacy. The need for effective privacy when it comes to media and communications has been brought into full focus in the wake of recent events such as Cambridge Analytica [6] and the privacy law suits raised against the Netflix recommender algorithms [39]. The international impact exercised by such concerning situations have prompted much dialogue around privacy as a socio-technical challenge. Here, increasing attention towards the role of the service providers (designers, researchers, content creators, developers etc.) to contribute towards privacy preserving processes within technologies is to be highlighted. At a governmental level, the GDPR response to such a perspective is it's Article 25, 'Data protection by design and default' [15]. On an ethical level, 'Privacy by Design' [8] identifies the scope of active design processes that contribute to the technology being consistent with the values of the EU Charter of Fundamental Rights [41]. HCI has been a key contributor to the realisation of such policies, often forming the bridge that connected these high level, generic agendas to design recommendations [22, 42], highlighting the need to go beyond just EULAs

[46] to engage in sustained, privacy preserving data management practices that engages and informs the users more [35].

Transparency and Control become another challenge that confronts the turn to personal data in media. There have been increasing calls for more transparency and control in data-driven media processes [33, 35] and the concern over transparency and control are constantly engaged with on ethical and policy levels [16]. While HCI response here is vast, recommendations that have inspired our intervention include the proposal for increased use of iconography [23], the effective leverage of timing in transactional notices [17, 36] and layering privacy notices to fit the context of the users by providing varying levels of information detail, to suit their interest [29].

Recent research has shown that lack of Accountability is a multi-dimensional challenge stemming from personal data use in media [34]. From a legal perspective it poses the need for both internal and external accountability [10]. This is closely followed by the need for computational accountability [10] which involves 'providing an account of' the underlying data practices which is of particular interest within media experiences, where black-boxing, filter bubbles and echo chambers are identified as ongoing concerns [32]. Accountability here also evokes responsibility to the public - social accountability. This involves being accountable as a societal entity, respecting and mitigating the social implications of using personal data that could otherwise lead to loss of user trust in the organization [27]. Social accountability holds a central position in relation to media as its scope also extends to service providers taking responsibility of the many potential 'intangible risks' [34] that arise from the inherent social nature of media consumption.

Finally, all the aforementioned challenges could impact Trust in media experiences. As media organisations start collecting personal data, their suitability as 'data brokers', the ethics of data use and the impact of data-driven processes on the audience experience have come to the fore [32]. Here, media organisations have put forth entire research agendas [4] to study and alleviate this lack of trust in the changing scape of the media-digital economy where HCI recommendations of "ubiquitous access and control of data. . .[. . .]. . . that help users manage personal data sharing" [35] are held at the forefront.

These challenges in media experiences are often treated separately, siloed within diverse disciplines, through varied theoretical traditions, methodologies and perspectives. Hence, there remains a gap in research around holistic, contextualised responses to these challenges in relation to emergent data driven media experiences. As media experiences become increasingly data driven, these socio-technical challenges take on urgency as they embed themselves deeper into the everyday lives of the audiences. This situation calls for future media experiences to be sensitive to these varied challenges and embody user centred design responses to their implications. Hence, we turn our focus to a user-oriented response framework "Human Data Interaction", inspired by HCI, which takes into account the multi-disciplinary nature of these discussed challenges and proposes holistic responses to them through an interdisciplinary approach that distills out three core principles.

3 HUMAN DATA INTERACTION

HDI is an interdisciplinary research area inspired by HCI that is focused on individuals in a data economy [11], their interactions with their data, the algorithms processing it and the services rendered by this processing. It draws together siloed learnings from different disciplines like law, technology, economics, ethics, psychology, sociology etc. [25] to provide three holistic yet focused response measures to the socio-technical challenges presented by data leverage - legibility, negotiability and agency.

3.1 Legibility

Legibility is defined as the process of making data interactions (what data is collected and/or processed), associated inferences and implications comprehensible. Whereas transparency [16] is limited to exposing what is happening, legibility goes a step further to prioritise understanding of that information. Stepping away from generic tenets of legibility like strong assertions of “the importance of providing information in relation to what the system is capable of (G1)” [2], recent research covers a wide range of issues around “intelligible intelligence” [47] and its human-centred HCI challenges, which highlight the social, practical and contextual nature of legibility, calling for it to be more like a conversation. Useable privacy is a prominent response to this call which often focuses on improving data legibility through “design aspects that can impact the effectiveness of privacy notices” [36] and Terms and Conditions statements. However, while such documents remain central in terms of need for increased legibility, there is the need to extend the view of HCI research to look at legibility in services beyond EULA documents to an on-going relationship that is in sync with the service itself. While recent HCI research has begun such exploration within the scope of AI [21] and even experimental media [33], the ‘fledgeling’ status of these attempts only further reinforces the significant potential for HCI intervention here.

3.2 Negotiability

For data-driven technologies, negotiations become a central theme when considering the challenge of privacy. Here, P3P (Platform for Privacy Preferences) [44] and APPEL (A P3P Preference Exchange Language) [45] are responses designed as tools for privacy policy preference expression. While these responses are steps towards providing increased negotiability within data driven systems, their effective realisation and applicability in the real world is yet to be realised. For example, the work by Yee and Korba [48] shows that understanding, beyond the application of tools like P3P, are required to enable active negotiations aimed at “eliminating or mitigating the unexpected bad outcomes” like loss of privacy. The work by George Yee and Larry Korba presents an approach for the negotiation of privacy policies for an e-learning service where negotiation “is done through human-mediated computer-assisted interaction rather than through autonomous agents.” [49] Thus, negotiation research begins to step away from just machine mediations to allow for more human intervention. While the examples here do not comprehensively exhaust literature around negotiations, they are used to represent some key stages of this research area. What this account shows is a promising space for contribution from HCI research that explores diverse avenues for user intervention and

participation in the process of data negotiability using user centred and design-oriented processes. It also highlights the focus of negotiability responses on privacy as a challenge, identifying the need for inclusion and response to the other challenges persistent in this scenario.

3.3 Agency

In 1999, Culnan and Armstrong defined control as giving the user “voice and control” over actual outcomes [13]. Hui et al. [18] views the scenario differently and suggest that control over personal data is restricted to the actions of the service provider organisation with little room for user agency once the information is disclosed. Thus shifting the focus onto who has the ability to act with this data in the world. To address this, Van Dyke et al. [14] introduce the notion of “privacy empowerment” where they report that “firms which meet the demand for control through empowering the user are rewarded with lower levels of privacy concern and increased trust”, thus presenting a compelling argument for shifting to a model that gives users more control over personal data. We build upon Van Dyke’s notion of ‘privacy empowerment’ advocating for increased HCI investment in methodologies that enable increased user participation through active control measures. Personal Data Stores (PDS) are a class of technology designed around this particular ethos which suggests a central role for the users in the personal data ecosystem by helping them maintain a more engaged relationship with their personal data. As most PDS-es leverage edge computing and client side processing, users retain control over data sharing, oversee who accesses their data, and have increased transparency around purposes of data use, thus giving the users a way to participate in the digital economy in a more active manner [43]. Examples of PDS enabling technologies include openPDS [24], Solid [40] and DataBox [1]. With this new range of edge technologies, HCI is presented with the unique opportunity of studying this alternative response, in terms of its applicability, useability and effectiveness in the everyday lives of the users.

The remainder of this paper reports the design of a novel media profiler service enabled by a Personal Data Store. This prototype provides a solid use case that speaks to the gap in media research around responses to the challenges identified while realising the opportunities for extending the HDI agenda, highlighted in the previous account. We choose a media profiler/ recommender as the vehicle for exploration of these gaps because the use of data in recommenders has specifically been shown to lead to concerns around trust, privacy [19], accountability transparency and control [35]. We leverage this contextual sensitivity to promote an effective space for enabling the application and study of responses to these challenges.

4 THE CROSS MEDIA PROFILER (CMP)

This research leverages a prototype of a new media service proposition that demands substantial amounts of user personal data while being sensitive towards this data collection and use through design and by default. This experience is a media content discovery service built on a personal data store that allows the user to ‘link’ (import copies of) their media profiles and associated data (eg., viewing history, playlists) from different digital media platforms like Spotify,



Figure 1: (a) Screen 1 : Introduction to the service and the concept of the PDS. (b) Screen 2 : A flow diagram showing the various stages of sign-up/ data remittance by the user. [Indicative designs for research purposes only] ©BBC R&D

Instagram and BBC. This allows them to create a combined media profile under the users' control to retrieve recommendations based on their media interests across different digital media platforms. This experience provided a strong use case for the instantiation of HDI principles because it demands a significantly high(er) amount of user personal data within a media experience, which helps highlight the data processes more explicitly to the users.

4.1 Design

The idea and execution of the CMP was led by an interdisciplinary team of industry experts (and academic partners) from a leading international public media organisation who contributed to various aspects of the research and development of the prototype as per their interest and expertise. The rest of this paper reports the design and future research this prototype is anticipated to enable.

The design initiated with the technological weaving together of the separate systems involved here (the personal data store and the APIs from various media platforms like Instagram, Spotify and BBC). This helped understand the capabilities, potential and limitations of the CMP prototype. Once the basic technological framework was put in place the prototype design was improved upon in an iterative manner by various sub-teams (eg., UX, design, HDI, technology, policy) within the interdisciplinary team to allow different wings to focus and improve upon their particular area of expertise, while allowing for critique and feedback from the wider team.

This process resulted in an onboarding flow website which was designed to help users understand the data sensitive affordances and novelties specifically designed into the CMP while orienting them towards its initial use and setup. The first 'complete' version of this website was used to conduct 4 expert walkthroughs where 4 experts from differing areas of interest (eg., UX, design, HDI, media research) experienced the flow and reported their feedback in a 'think-aloud' manner.

This formative research effort highlighted areas of improvement for the website flow, particularly around language disambiguation, number of screens between the initial and final setup pages, amount of information on certain key screens, improved use of iconography, highlighting of key concepts and options for more choice and control. This information was fed into subsequent design iterations

to revise the previous prototype version to include these recommendations and result in the following 6 key screens that would be presented to audiences.

4.2 Inclusion of HDI principles into Design

The research reported in the rest of this paper focuses on and explicates aspects of this design that concerns the research agenda stated previously : the realisation of the principles of HDI as a response to the socio-technical challenges confronting personal data use in media experiences. As part of this agenda, the principles of legibility, negotiability and agency were included into the design of the CMP's conceptualisation and the resulting onboarding flow design. The following section explains how the three theoretical principles are practically integrated into the design of the prototype, while citing wider research that supported the specific design decisions wherever applicable.

4.2.1 Legibility. Data legibility is designed into different levels in the CMP. Firstly, the language used during the signup process was consciously simplified (with help from a professional copywriter) to be simple and accessible [7] (Screens 1 and 4). Secondly, information about the data collection, linking and the resulting recommendations were distributed over different stages of the signup, using flowcharts, icons, diagrams, hierarchical information delivery - through buttons and links - and easily noticeable text pointers (Screens 1, 2 and 3) [23, 29, 36]. Thirdly, although the full T&Cs of the service was available to the users, a permissions page was designed to communicate key points about the service to the user including data collection and data linking in a quick and simple manner with minimal text (Screen 4) [29].

4.2.2 Negotiability. Data negotiability was built into the core of the CMP through the notion of 'data linking'. Instead of being forced to sign up through their various accounts, users are given the choice to import their data from different media platforms to improve their recommendations on this service (Screen 3). In doing so they are given the opportunity to make active and informed choices [5, 12] around which media platforms (eg., Spotify, Instagram) they connect to.

4.2.3 Agency. Agency with regard to personal data is enabled through the Personal Data Store [1]. Users have the ability to review

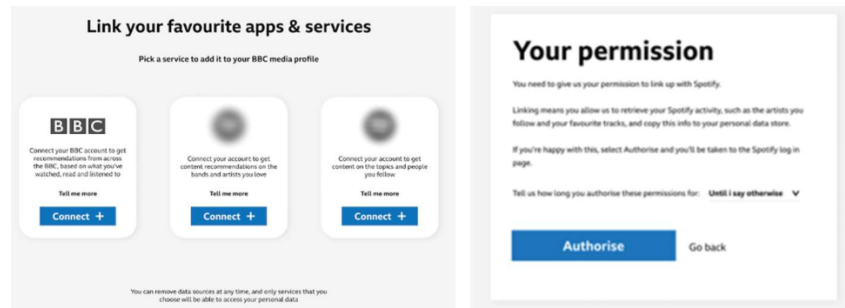


Figure 2: (a) Screen 3 : The data linking page that allows the user to choose which applications to connect to and import data from, for linking data. (b) Screen 4 : The permissions page that summarises the data flows being consented to. [Indicative designs for research purposes only] ©BBC R&D

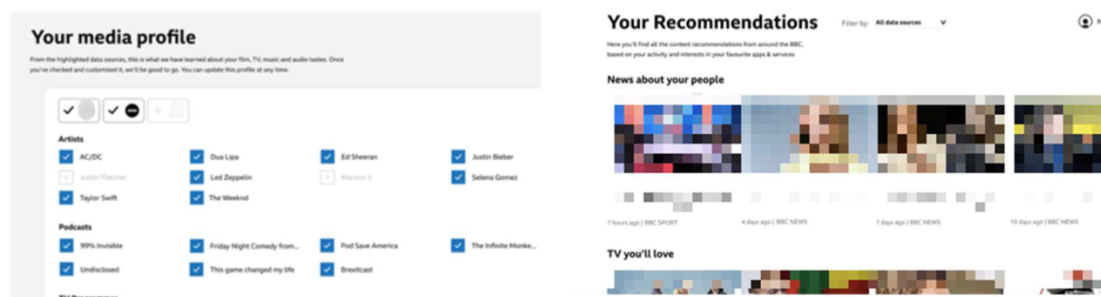


Figure 3: (a) Screen 5 : The media profile that allows for granular data control. (b) Screen 6 : The resulting sample recommendations page. [Indicative designs for research purposes only] ©BBC R&D

and make edits to this data through a media profile interface that presents the data they have imported from various different digital services (Screen 5). This PDS ensures that all data that is linked into the system stays client-side, at the edge of the network, where the processing is conducted and results delivered through the CMP (Screen 6). Here, the users are able to filter their recommendations, based on data sources eg., choosing to view recommendations based on all the data they have linked or choosing a subset (one or more eg., Instagram and/ or Spotify and/or BBC) of the sources they have chosen to link with this account, thus being able to see the ‘effect’ of their data interactions, which makes engagement with their personal data more relatable [33].

5 FUTURE RESEARCH

The reported design of the CMP, a lightweight onboarding website flow introducing the concept of ‘data linking’, the use of the PDS, the benefits of this novelty and the options for enhanced user negotiability and agency over their data would be used as a research probe for formative studies around this service proposition and accompanying data sensitive responses. The formative studies, which would be a set of online focus groups with general audiences would be used for feedback around the responses built into the system, the legibility offered by the onboarding flow about these responses and the overall concepts put to practice here. The responses would be used to improve upon this current version of the prototype before taking it into a wider audience trial which

would have audiences experiencing the service for a 10 day period, thus extending their engagement beyond just the onboarding and the concepts to an experience of ‘living’ with this service. This trial would be used as an evaluative study employing a mix of qualitative and quantitative research methods that study the effectiveness (or lack thereof) of the design decisions made around data legibility, negotiability and agency as a means for alleviating the socio-technical challenges of personal data leverage in future media experiences, thereby identifying and producing practical design recommendations and patterns that could be adopted by both researchers and practitioners working within this scope.

6 SELF DISCLOSURE

Upon recognition of the fact that we, as researchers, influence the research and its outcomes, we make our backgrounds and motivations explicit here. The authors of this paper are residents of the United Kingdom and of White and Asian Indian ethnicity. All members associated with the design of this prototype (including the wider interdisciplinary team) are residents within the U.K. and English speakers. Hence, our notion of legibility is currently constrained to resources designed and delivered in English. The project looks into the legal, social and ethical implications of personal data use and at this stage of the research these considerations are reflective of practices prevalent and applicable to the U.K.. We understand that challenges around personal data, particularly legal mandates could often be country-specific and we accept this as a limitation

of our current approach. We also understand the varied spectrum of subtleties associated with social and ethical challenges, when considering different cultures, languages and geographic locations, wherein we use this prototype as an instantiation representing the specific yet diverse legal, social and ethical nuances of the people and society we and our technologies are immediately exposed to. We consider this focus important to unpack the underlying socio-technical sensitivities within our study context and hope for future research to extend upon it by either proving it as generalisable to other contexts or by building upon it. But, with the view of producing results that are representative of a global audience we intend to engage with study participants from diverse backgrounds and priorities (within the constraints stated above), with an explicit focus of maximized inclusion of various different audience cohorts (in terms of age, gender, ethnicity, educational background, professional status etc.).

7 CONCLUSION

The turn to personal data leverage by media experiences has introduced a number of socio-technical challenges like concerns around privacy, transparency, control, accountability and trust, which confront the adoption of these novel technologies. While these phenomena have been studied in various disciplines within industry and academia alike, practical solutions that effectively respond to them are still missing. This calls for holistic, inter-disciplinary response measures that are capable of speaking to the various dimensions of these wider challenges simultaneously. Hence, we look at Human Data Interaction, an emergent multi-disciplinary approach inspired by HCI that aims to respond to said challenges through three overarching principles : data legibility, negotiability and agency. This paper reports and rationalizes the practical integration of HDI principles within a Cross Media Profiler : a novel media service that combines user personal data from different digital media platforms to provide more holistic content recommendations. This effort, while aimed at responding to the discussed challenges within media experiences through the application of HDI, also furthers the current interests of the HDI agenda by realizing the theoretical tenets of HDI in a practical manner.

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